

The Economics of Bicycles for the Mind

Cognitive tools, human judgment, and inequality

Gabriel Saco

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Universidad del Pacífico

Agrawal, Gans & Goldfarb (2025) · NBER WP 34034

github.com/gsaco/ai-02-agrawal

Extended technical presentation

Roadmap: model, propositions, and audit

1. Mechanism

A cognitive tool raises direct performance but lowers the marginal return to implementation effort.

2. Propositions

Less effort, more value; adoption depends differently on skill and two forms of judgment.

3. Inequality audit

Identify the variance object, derive its threshold, and check the paper's endpoint claim.

Contribution of this repository

A symbolic derivation, an exact-moment counterexample, a separate individual-benefit calculation, and a documented interior-domain audit.

Motivating example I: computer-aided design

Without the tool

The worker calculates and draws manually. Each improvement uses implementation effort e , transformed by skill s into output $v(se)$.

$$M(e) = v(se) - c(e),$$
$$v'(se^*)s = c'(e^*).$$



With the tool

Software implements calculations and visualizations more easily. Direct task value rises while additional human effort becomes less useful.

$$e^*(1) < e^*(0),$$
$$M(e^*(1); 1) > M(e^*(0); 0).$$

Computer assistance is the paper's clean example of "more output with less input."

Motivating example II: AI-assisted prediction

Diagnosis separates prediction from judgment

Effort and implementation skill improve diagnostic accuracy $p(se)$. Payoff judgment α is the ability to select and apply the right treatment after receiving a signal.

Objective

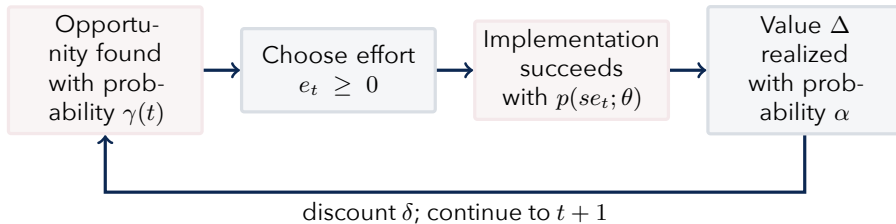
$$\max_{e \geq 0} \alpha \{p(se)\Delta_C + [1 - p(se)]\Delta_W\} - c(e).$$

AI shifts accuracy from $p(se)$ to $p(se + \theta)$.

Economic separation

- ▶ Better prediction is valuable only when the decision-maker can act.
- ▶ Endogenous effort may fall after tool adoption.
- ▶ Payoff judgment complements the tool only if realized accuracy rises.

The task-improvement process



Stopping rule

If no new opportunity is perceived, the process ends. Conditional on an opportunity, the effort problem is identical in every round.

Three human capabilities

Implementation skill s

How effectively effort becomes a successful implementation. It enters through se_t .

Payoff judgment α

The probability of recognizing and extracting value from a successful implementation.

Opportunity judgment $\gamma(t)$

The probability of finding another opportunity to improve the task in round t .

Why the decomposition matters

A tool can substitute for implementation skill while amplifying the value of judgment. Treating all three as a single "skill" would hide the distributional mechanism.

General optimization problem

Per opportunity

$$e_t^*(\theta) \in \arg \max_{e_t \geq 0} \{p(se_t; \theta)\alpha\Delta - c(e_t; \theta)\}.$$

Interior FOC:

$$p'(se_t^*; \theta)s\alpha\Delta = c'(e_t^*; \theta).$$

Across opportunities

$$V_0(\theta) = \sum_{t=0}^{\infty} \left(\prod_{i=0}^t \gamma(i) \right) \delta^t M(e_t^*(\theta); \theta).$$

The opportunity path scales value, but it does not change the conditional effort problem.

Definition 1: what counts as a cognitive tool?

Direct effects

For every e and $\theta' > \theta$,

$$p(se; \theta') \geq p(se; \theta), \quad c(e; \theta') \leq c(e; \theta),$$

with a strict improvement somewhere relevant for a strict value gain.

Marginal substitution

For $e > 0$,

$$\frac{p'(se; \theta')}{c'(e; \theta')} < \frac{p'(se; \theta)}{c'(e; \theta)}.$$

Higher tool quality reduces the marginal success obtained per unit of marginal human effort cost.

Proposition 1: more value with less effort

Statement

As tool quality moves from $\theta = 0$ to $\theta = 1$,

$$e_t^*(1) < e_t^*(0) \forall t, \quad e_t^*(\theta) = e^*(\theta) \forall t, \quad V_0(1) > V_0(0).$$

Conditions carried by the result

A well-defined selected optimum; the cognitive-tool inequalities; positive $s\alpha\Delta$ and finite discounted opportunities; and interior optima or the corresponding strict Kuhn-Tucker comparison at the boundary.

The time-invariance result comes from repeating the same conditional problem, not from constant opportunity arrival.

Proposition 1: proof logic and economics

Effort comparison

The FOC equates $s\alpha\Delta p'/c'$ to one. Definition 1 shifts this ratio downward at every positive effort, so the new optimum moves to a lower e under the paper's monotonicity logic.

Value comparison

At the old effort, the tool weakly raises success and weakly lowers cost. Re-optimizing cannot reduce that improved payoff:

$$M(e^*(1); 1) > M(e^*(0); 0).$$

Economic intuition

The tool handles the low-hanging implementation work. Human effort falls, but optimized net task value rises because the entire opportunity payoff schedule has improved.

Adoption value and the opportunity multiplier

Factorization

Define

$$\Gamma = \sum_{t=0}^{\infty} \left(\prod_{i=0}^t \gamma^{(i)} \right) \delta^t.$$

Since optimal effort is time-invariant,

$$V_0(1) - V_0(0) = \Gamma [M(e^*(1); 1) - M(e^*(0); 0)] > 0.$$

Interpretation

Adoption value equals a per-opportunity productivity gain multiplied by the discounted number and timing of opportunities on which the tool can be used.

Proposition 2.1: opportunity judgment

General sequence

Every component $\gamma(t)$ affects the probability of reaching all later opportunities. Hence better opportunity judgment raises adoption value whenever the per-opportunity gain is positive.

Geometric special case

If $\gamma(0) = \gamma_0$ and $\gamma(t) = \gamma$ for $t > 0$,

$$\Gamma = \frac{\gamma_0}{1 - \delta\gamma}, \quad \delta\gamma < 1.$$

Mechanism

Opportunity judgment does not make implementation better at one opportunity; it increases how often the agent can apply the tool's gain.

Proposition 2.2: payoff judgment

Envelope calculation

$$\frac{\partial[V_0(1) - V_0(0)]}{\partial\alpha} = \Gamma\Delta [p(se^*(1); 1) - p(se^*(0); 0)].$$

Complementarity

Payoff judgment raises adoption value if realized implementation success is weakly higher with the tool.

Why the sign is ambiguous

The direct tool effect raises success, but endogenous effort falls. Definition 1 alone does not determine which force dominates.

Proposition 2.3: implementation skill

Optimizer-level derivative

$$\frac{\partial[V_0(1) - V_0(0)]}{\partial s} = \Gamma\alpha\Delta\{p_s(se^*(1); 1) - p_s(se^*(0); 0)\}.$$

Paper's condition and the caveat

The paper invokes $p_{s\theta} < 0$ to call tool quality and skill substitutes. But that cross-partial holds effort fixed, whereas the displayed derivative compares two different optimal efforts. An additional restriction is required to sign the optimizer-level bracket in full generality.

The intended mechanism remains inverse skill bias: lower-skill workers receive the larger marginal tool boost.

Proposition 2.4: early versus late opportunities

Correct derivative for a general path

$$\frac{\partial \Gamma}{\partial \gamma(t)} = \sum_{k=t}^{\infty} \delta^k \prod_{\substack{i=0 \\ i \neq t}}^k \gamma(i).$$

Changing $\gamma(t)$ affects the chance of reaching every subsequent $k \geq t$.

Audit of the displayed proposition

Proposition 2's ratio keeps only the first $k = t$ product. The general ranking therefore depends on the whole opportunity path. Earlier opportunities do receive greater weight in the geometric special case under discounting.

Inequality specialization: all maintained assumptions

Functional form

$$p(se; \theta) = \sqrt{se + \theta}, \quad c(e) = e.$$

$\gamma(0) = \gamma_0$, $\gamma(t) = \gamma$ for $t > 0$, and $\delta\gamma < 1$.

Tool quality

$\theta \in [0, \infty)$ is treated as continuous, even though the earlier adoption comparison used $\theta \in \{0, 1\}$.

Heterogeneity

$\alpha, \gamma_0, \gamma, s$ are independent, have positive support, means μ_i , variances σ_i^2 , and satisfy the paper's $\mu_i > 3\sigma_i$ restriction.

Interiority

The closed form applies only where every worker satisfies

$$\theta < \alpha^2 \Delta^2 s^2 / 4.$$

Closed form: effort, net benefit, and value

Solve the FOC

$$\frac{\alpha \Delta s}{2\sqrt{se^*(\theta) + \theta}} = 1 \quad \Rightarrow \quad e^*(\theta) = \frac{\alpha^2 \Delta^2 s}{4} - \frac{\theta}{s}.$$

Per opportunity

$$M(\theta) = \frac{\alpha^2 \Delta^2 s}{4} + \frac{\theta}{s}.$$

Tool quality has an inverse-skill boost θ/s .

Continuation value

$$V(\theta) = \Gamma \left(\frac{\alpha^2 \Delta^2 s}{4} + \frac{\theta}{s} \right).$$

Opportunity judgment multiplies both components.

Proposition 3: corrected exact statement

Mean

$$\frac{d\mathbb{E}[V(\theta)]}{d\theta} = \mathbb{E}[\Gamma]\mathbb{E}[1/s] > 0.$$

Variance

If

$$\frac{\mathbb{E}[\Gamma^2]}{\mathbb{E}[\Gamma]^2} < \mu_s \mathbb{E}[1/s] \quad \text{and} \quad \text{Var}(\Gamma/s) > 0,$$

then the interior $\text{Var}[V(\theta)]$ is strictly convex, decreases before θ^* , and increases after θ^* .

Endpoint

The printed claim of a positive slope at $\theta = 1$ additionally requires $\theta^* < 1$ and an interior solution there.

Mean effect: productivity always rises

Expectation

Independence implies

$$\mathbb{E}[V(\theta)] = \mathbb{E}[\Gamma] \left\{ \frac{\Delta^2}{4} \mathbb{E}[\alpha^2] \mu_s + \theta \mathbb{E}[1/s] \right\}.$$

Comparative static

$$\frac{d\mathbb{E}[V(\theta)]}{d\theta} = \mathbb{E}[\Gamma] \mathbb{E}[1/s] > 0.$$

Every worker's interior tool boost is positive, and averaging preserves the sign.

The controversy is about dispersion, not the sign of mean productivity.

Variance algebra I: separate multiplier and payoff

Product variance

Since Γ and $M(\theta)$ are independent,

$$\text{Var}[V(\theta)] = \mathbb{E}[\Gamma^2] \text{Var}[M(\theta)] + \text{Var}(\Gamma) \mathbb{E}[M(\theta)]^2.$$

Quadratic per-opportunity variance

With $K = \Delta^2/4$,

$$\mathbb{E}[M] = K\mathbb{E}[\alpha^2]\mu_s + \theta\mathbb{E}[1/s], \quad \text{Var}(M) = A + B\theta + C\theta^2,$$

where $B = 2K\mathbb{E}[\alpha^2]\{1 - \mu_s\mathbb{E}[1/s]\} \leq 0$ and $C = \text{Var}(1/s) \geq 0$.

Linear slope

Collecting the derivative gives

$$\begin{aligned}\frac{d \operatorname{Var}[V(\theta)]}{d\theta} &= a_0 + 2\theta \operatorname{Var}(\Gamma/s), \\ a_0 &= \frac{\Delta^2 \mathbb{E}[\alpha^2]}{2} \{ \mathbb{E}[\Gamma^2] - \mathbb{E}[\Gamma]^2 \mu_s \mathbb{E}[1/s] \}.\end{aligned}$$

Condition (30)

$$\frac{\mathbb{E}[\Gamma^2]}{\mathbb{E}[\Gamma]^2} < \mu_s \mathbb{E}[1/s] \iff a_0 < 0.$$

It is precisely the condition for an initially negative variance slope.

Turning point: what the U-shape really says

Unique interior minimum

If $\text{Var}(\Gamma/s) > 0$,

$$\theta^* = -\frac{a_0}{2 \text{Var}(\Gamma/s)} > 0,$$

with a negative slope below θ^* and a positive slope above it.

Two competing forces

Inverse skill bias: θ/s helps low- s workers more.

Judgment amplification: heterogeneous Γ magnifies who can use that boost repeatedly.

Correct scope

This is a result for the cross section and for the common interior tool-quality range. It is not a global claim over $\theta \in [0, \infty)$.

The professor's three questions, answered

1. Variance of what, with respect to what?

Cross-sectional variance of continuation value $V_i(\theta)$, interpreted as wages, as continuous tool quality θ changes.

2. Is the result unconditional?

No. It needs the specialized functional form, independence and support restrictions, $\delta\gamma < 1$, condition (30), $\text{Var}(\Gamma/s) > 0$, and a common interior range.

3. What about individual benefit?

$$D_i(\theta) = \theta \Gamma_i / s_i, \quad \text{Var}[D(\theta)] = \theta^2 \text{Var}(\Gamma/s),$$

so benefit variance rises monotonically for $\theta > 0$.

Variance section

- ▶ Equation (32) needs $\theta^* < 1$ for its slope-at-one claim.
- ▶ The printed $\Delta > 2/(s\sqrt{\alpha})$ does not ensure $e^*(1) > 0$; the algebra requires $\Delta > 2/(\alpha s)$.
- ▶ That corrected interior condition forces $p^*(1) > 1$, conflicting with the general model's probability bound.

Proposition 2

- ▶ The displayed general timing ratio omits all future terms affected by $\gamma(t)$.
- ▶ $p_{s\theta} < 0$ holds effort fixed and does not alone sign the comparison across two endogenous effort choices.

These points qualify particular claims; they do not erase the model's central substitution mechanism.

Exact-moment counterexample and SymPy audit

Independent-uniform construction

All variables are mutually independent, with

$$\alpha \in [.75, .85], \quad \gamma_0 \in [.48, .52], \quad \gamma \in [.36, .44], \quad s \in [.8, 1],$$

and $\delta = .8$, $\Delta = 5$. Their supports are positive and each satisfies $\mu_i > 3\sigma_i$.

Exact symbolic and moment checks

$$\frac{\mathbb{E}[\Gamma^2]}{\mathbb{E}[\Gamma]^2} = 1.0012732 < 1.0041460 = \mu_s \mathbb{E}[1/s], \quad \theta^* = 1.7007141,$$
$$\left. \frac{d \text{Var}[V(\theta)]}{d\theta} \right|_{\theta=1} = -0.0051337 < 0, \quad \min_i \frac{\alpha_i^2 \Delta^2 s_i^2}{4} = 2.25.$$

What the counterexample establishes

Condition (30) holds and the turning point lies in the common interior range, yet the variance slope is still negative at $\theta = 1$. Thus the paper's endpoint claim needs the missing condition $\theta^* < 1$.

Limiting cases and final takeaways

Limiting cases

- ▶ Constant Γ : skill heterogeneity can generate the initial compression.
- ▶ Constant s : the equalizing inverse-skill channel disappears.
- ▶ Constant Γ/s : no strict quadratic curvature.
- ▶ Large θ : workers hit $e = 0$ at heterogeneous thresholds.

Three takeaways

1. Cognitive tools can raise value while reducing human effort.
2. Skill and two forms of judgment have different adoption returns.
3. The U-shape is conditional, local to the interior regime, and is not the variance of individual adoption benefits.

Bottom line

The paper's mechanism is valuable; the exact algebra makes its empirical and inequality claims sharper than the loose "AI is U-shaped" slogan.